User Manual

Team 09

2252709 Xuanhe Yang
2251548 Jingxiao Han
2251653 Tong Li
2253715 Fubin Chen

Abstract

This comprehensive user manual for the Smart Maintenance Platform for Aero Engine Industrial Equipment (Proactive Shield) provides detailed guidance on all system functionalities and operational procedures. Designed for various user roles including operators, maintenance engineers, managers, and system administrators, the manual covers the platform's deep learning-based predictive maintenance capabilities, including anomaly detection using Skip-GANomaly and remaining useful life prediction using CNN-LSTM models. The document details the system's core modules—Device Center, Monitoring Center, Data Simulation, Alert System, and Reporting System—along with advanced features like maintenance schedule optimization. Complete with best practices, troubleshooting guidelines, and reference materials, this manual serves as the definitive resource for effectively utilizing the platform to achieve up to 30% reduction in unplanned downtime and 25% decrease in maintenance costs through data-driven maintenance decision support.

Team Introduction

This project was collaboratively completed by the following team members:

Xuanhe Yang
Student ID: 2252709
Phone: +86 18937082731
Email: 2252709@tongji.edu.cn

Jingxiao Han
Student ID: 2251548
Phone: +86 13817024314
Email: 2251548@tongji.edu.cn

Tong Li
Student ID: 2251653
Phone: +86 18545556320
Email: 2251653@tongji.edu.cn

Fubin Chen
Student ID: 2253715
Phone: +86 15159581396
Email: 1958440015@qq.com

Table of Contents

Chapter1 Document Information

1 Document Control Information

Document Title	Smart Maintenance Platform for Aero Engine Industrial Equipment - User Manual
Version	1.5 (Final Draft)
Date	May 15, 2025
Status	Completed
Document Owner	Project Team

2 Revision History

Version	**Date**	**Description**	**Author**
0.1	April 2, 2025	Initial structure	Tong Li
0.5	April 4, 2025	Core functionality overview	Jingxiao Han
1.0	April 6, 2025	Completed initial draft	Project Team
1.3	April 28, 2025	Updated with core functionality details	Tong Li
1.5	May 15, 2025	Completed frontend implementation and UI documentation	Fubin Chen, Tong Li

3 Approvals

Name	**Role**	**Signature**	**Date**
Xuanhe Yang	Project Manager	YANGXUANHE	May 16, 2025
Jingxiao Han	Team Member	HANJINGXIAO	May 16, 2025
Tong Li	Team Member	LITONG	May 16, 2025
Fubin Chen	Team Member	CHENFUBIN	May 16, 2025

4 Document Scope

This user manual provides comprehensive guidance for operators, maintenance personnel, and administrators of the Smart Maintenance Platform for Aero Engine Industrial Equipment. It covers detailed descriptions, operational procedures, and best practices for all functional modules of the system, aimed at helping users fully utilize the platform's predictive maintenance capabilities.

Chapter2 Introduction

1 System Overview

The Smart Maintenance Platform for Aero Engine Industrial Equipment (Proactive Shield) is a deep learning-based predictive maintenance system designed to improve aero engine reliability, reduce unexpected downtime, and optimize maintenance schedules. The system works by collecting and analyzing sensor data to detect potential anomalies and predict the remaining useful life (RUL) of critical components.

The platform integrates advanced anomaly detection algorithms (Skip-GANomaly, PHOT) and life prediction models (CNN-LSTM, CNN-Transformer, DBN-BiGRU) to provide data-driven support for maintenance decisions. The system presents complex analytical results through an intuitive user interface, enabling technicians to make informed maintenance decisions.

Key system objectives include:

- Reducing unplanned downtime by 25-30% through predictive maintenance
- Decreasing maintenance costs by 20-25%
- Improving Overall Equipment Effectiveness (OEE) by 15 percentage points
- Detecting anomalies with greater than 95% accuracy
- Providing accurate Remaining Useful Life (RUL) predictions for critical components

2 Target Audience

This user manual is intended for the following user groups:

- **Operators:** Technical personnel responsible for daily monitoring and basic system operations
- **Maintenance Engineers:** Professionals responsible for analyzing diagnostic results and performing maintenance tasks
- **Managers:** Decision-makers who need access to reports and performance metrics
- **System Administrators:** IT personnel responsible for system configuration, user management, and permission settings

3 System Requirements

2.3.1 Hardware Requirements

Component	**Requirement**
Server	
Processor	Minimum: Intel Xeon E5-2650 v4 or equivalent Recommended: Intel Xeon Gold 6248R or equivalent
Memory	Minimum: 32GB RAM Recommended: 64GB RAM or higher

Storage	Minimum: 1TB SSD Recommended: 2TB NVMe SSD + 4TB HDD for data storage
Network	1Gbps Ethernet (minimum) 10Gbps Ethernet (recommended)
Client	
Processor	Minimum: Intel Core i5 (8th gen) or equivalent Recommended: Intel Core i7 (10th gen) or equivalent
Memory	Minimum: 8GB RAM Recommended: 16GB RAM
Display	Minimum: 1920 x 1080 resolution Recommended: 2560 x 1440 resolution or higher
Network	Minimum: 10Mbps connection Recommended: 100Mbps connection

2.3.2 Software Requirements

Component	**Requirement**
Server	
Operating System	Ubuntu Server 22.04 LTS or later Red Hat Enterprise Linux 8 or later Windows Server 2019 or later
Database	MySQL 8.0 or later InfluxDB 2.0 or later (for time-series data)
Container Platform	Docker 20.10 or later Kubernetes 1.20 or later (for distributed deployment)
Client	
Operating System	Windows 10/11 macOS Monterey or later Ubuntu Desktop 20.04 or later
Web Browser	Google Chrome (latest 2 versions) Mozilla Firefox (latest 2 versions) Microsoft Edge (latest 2 versions) Safari (latest 2 versions)
Additional Software	PDF Reader for reports Spreadsheet application for exported data

2.3.3 Network Requirements

Requirement	**Specification**
Minimum Bandwidth	10Mbps per concurrent user
Recommended Bandwidth	100Mbps for the organization
Latency	¡100ms between client and server
Firewall Configuration	Outbound ports: 80 (HTTP), 443 (HTTPS) Inbound ports (server): 80, 443, 22 (SSH)
Security Protocol	TLS 1.2 or higher

4 Conventions and Terminology

2.4.1 Document Conventions

This manual uses the following formatting conventions:

- **Bold text:** Indicates menus, buttons, or interface elements
- *Italic text:* Indicates specially emphasized information
- `Code font`: Indicates system messages or input values

Note: Information that requires special attention appears in note boxes like this one.

Warning: Critical information that could prevent data loss or system problems appears in warning boxes like this one.

2.4.2 Glossary

Term	**Definition**
RUL	Remaining Useful Life, the estimated time a component can continue to operate before failure
PdM	Predictive Maintenance, using data analytics and machine learning to predict when equipment will need maintenance
Anomaly Detection	Identifying behaviors or states that deviate from normal operation patterns
Threshold	A predefined value that triggers an alert or notification
Alert	A system-generated reminder indicating a condition that requires attention
Deep Learning	A branch of artificial intelligence that uses multi-layered neural networks for pattern recognition and prediction
KPI	Key Performance Indicator, used to measure the performance of a system or component
CNN	Convolutional Neural Network, a deep learning algorithm suitable for image and pattern recognition
LSTM	Long Short-Term Memory, a type of recurrent neural network suitable for time-series data
GAN	Generative Adversarial Network, a deep learning technique used for anomaly detection
Skip-GANomaly	A specific GAN architecture that uses skip connections for improved anomaly detection
Transformer	A neural network architecture based on self-attention mechanisms
OEE	Overall Equipment Effectiveness, a measure of manufacturing productivity

Chapter3 Getting Started

1 Accessing the System

To access the Smart Maintenance Platform for Aero Engine Industrial Equipment:

1. Open a supported web browser
2. Navigate to the system URL: `https://proactive-shield.example.com`
3. Enter your username and password on the login page
4. Click the **Login** button

Note: On first login, you may be required to change your default password and set up security questions for account recovery purposes. Your administrator will provide your initial login credentials.

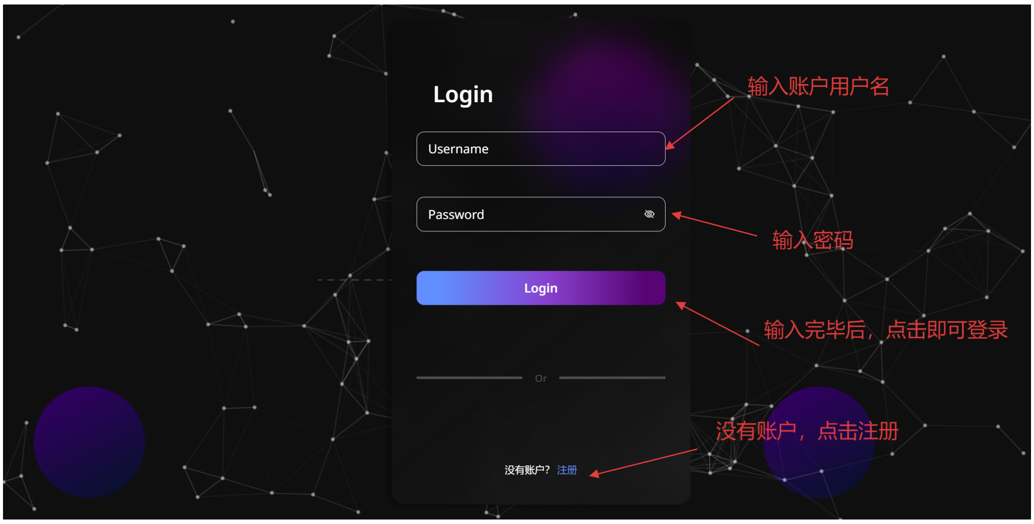

Figure 3.1: System Login Screen

2 User Interface Overview

The user interface includes the following main areas:

3.2.1 Navigation Menu

Located on the top of the interface, providing access to all main functions of the system:

- **Dashboard:** System overview and key metrics
- **Device Center:** Manage and monitor all engine equipment
- **Monitoring Center:** Real-time data and trend monitoring
- **Alert Center:** View and manage system alerts
- **Data Analysis:** In-depth analysis and reporting
- **System Administration:** Configure and manage system settings

3.2.2 Main Content Area

Occupies the central part of the interface, displaying the content of the currently selected function. Content changes dynamically based on the selected menu item.

Figure 3.2: Main User Interface Dashboard

3 Login and Logout

3.3.1 Logging Into the System

1. Navigate to the system login page at `https://proactive-shield.example.com`
2. Enter your username in the **Username** field
3. Enter your password in the **Password** field
4. Click the **Login** button

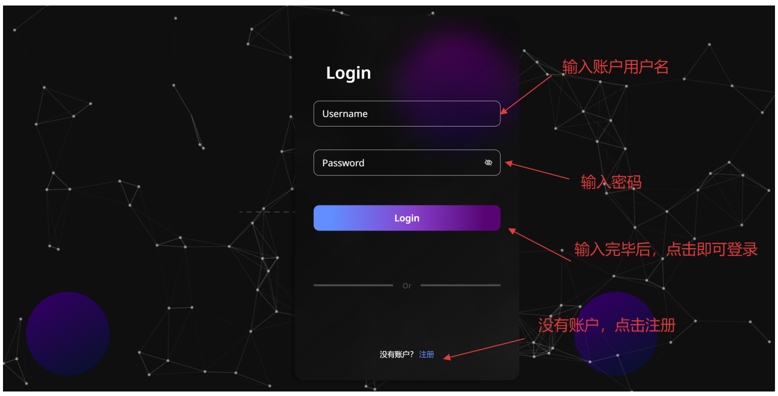

Figure 3.3: Login Page

3.3.2 Logging Out of the System

1. Click on the user icon in the top-right corner of the interface
2. Select **Logout** from the dropdown menu
3. The system will return to the login page

3.3.3 Password Reset

If you've forgotten your password:

1. Click the **Forgot Password** link on the login page
2. Enter your username or registered email
3. Follow the instructions sent to your email to reset your password

4 User Preferences

You can customize the system interface according to your personal needs:

1. Click on the user icon in the top-right corner of the interface
2. Select **Preferences**
3. Adjust the following options:
   - Interface language
   - Time zone settings
   - Notification preferences
   - Dashboard layout
   - Data display units
   - Color theme
4. Click **Save** to apply your changes

Chapter4　Core Functionality

1　Device Center

4.1.1　Overview

The Device Center is the central location for managing all monitored aero engines and components. It provides device registration, configuration, grouping, and status monitoring capabilities, enabling users to manage the entire device lifecycle.

Figure 4.1: Device Center Interface

4.1.2　Key Features

- Device registration and configuration
- Device status monitoring
- Device grouping and categorization
- Device details viewing
- Device maintenance history
- Device documentation management

4.1.3　Device Registration

To register a new device:

1. Navigate to the **Device Center**
2. Click the **Add Device** button
3. Complete the registration form with the required information:
    - Device name

- Device type
- Serial number
- Location
- Installation date
- Associated sensors

4. Click **Submit** to register the device

4.1.4 Viewing Device Details

To view detailed information about a device:

1. Navigate to the **Device Center**
2. Locate the device in the list or use the search function
3. Click on the device name or the **View Details** button
4. The system will display the device details page with tabs for:

- General Information
- Performance Metrics
- Maintenance History
- Associated Sensors
- Documentation

2 Monitoring Center

4.2.1 Overview

The Monitoring Center provides real-time and historical monitoring of engine performance. Users can view key parameters, analyze performance trends, and identify potential issues. This module utilizes advanced visualization techniques to make complex data easy to understand and interpret.

Figure 4.2: Monitoring Center Interface

4.2.2 Key Features

- Real-time data monitoring
- Historical trend analysis
- Multi-parameter correlation analysis
- Custom dashboards
- Threshold configuration and monitoring
- Data export and reporting

4.2.3 Real-time Monitoring

To monitor real-time data:

1. Navigate to the **Monitoring Center**
2. Select the device or component you want to monitor
3. View the current parameters displayed in the dashboard
4. Use the refresh rate selector to adjust the update frequency
5. Configure thresholds for parameters by clicking the **Configure Thresholds** button

4.2.4 Historical Data Analysis

To analyze historical data:

1. Navigate to the **Monitoring Center**
2. Select the device or component for analysis
3. Click the **Historical Data** tab
4. Set the date and time range for analysis
5. Select the parameters you want to analyze
6. Click **Apply** to generate the historical data visualization
7. Use the zoom and pan tools to explore the data

3 Data Simulation

4.3.1 Overview

The Data Simulation module allows users to simulate engine performance under various operating conditions. This is valuable for training purposes, predictive scenario analysis, and hypothesis testing. Users can configure different parameters and observe the simulated results.

Figure 4.3: Data Simulation Interface

4.3.2 Key Features

- Simulation scenario creation
- Parameter configuration
- Simulation run control
- Results visualization
- Comparison with historical data
- Scenario saving and sharing

4.3.3 Creating a Simulation

To create a new simulation:

1. Navigate to the **Data Simulation** module
2. Click **New Simulation**
3. Select the device or component type to simulate
4. Configure the operational parameters:
   - Operating conditions
   - Environmental factors
   - Time duration
   - Initial component conditions
5. Click **Start Simulation** to begin the simulation process

4 Alert System

4.4.1 Overview

The Alert System monitors engine performance and system predictions, generating notifications when anomalies are detected or potential issues are predicted. Users can configure alert rules, manage alerts, and track resolution processes.

Figure 4.4: Alert System Interface

4.4.2 Key Features

- Alert rule configuration
- Real-time alert monitoring
- Alert notifications (email, SMS, system notifications)
- Alert acknowledgment and handling
- Alert history recording
- Alert statistics and trend analysis

4.4.3 Viewing Alerts

To view current alerts:

1. Navigate to the **Alert Center**
2. The system displays a list of active alerts, sorted by severity
3. Use the filters to refine the alert list by:
   - Severity
   - Time period
   - Device type
   - Alert type
   - Status (acknowledged, unacknowledged)
4. Click on an alert to view its details

4.4.4 Managing Alerts

To manage an alert:

1. Click on the alert in the alert list
2. View the detailed information about the alert
3. Choose one of the following actions:

- **Acknowledge:** Indicates you are aware of the alert
- **Assign:** Assign the alert to a specific user
- **Resolve:** Mark the alert as resolved
- **Escalate:** Escalate the alert to a higher priority
- **Snooze:** Temporarily silence the alert

4. Add comments or notes as needed
5. Click **Save** to apply your changes

5 Reporting System

4.5.1 Overview

The Reporting System provides comprehensive reporting and analysis capabilities to help users understand equipment performance, track maintenance activities, and support decision-making processes. The system includes predefined reports and custom reporting options.

Figure 4.5: Reporting System Interface

4.5.2 Key Features

- Predefined reports (operational status, maintenance history, performance trends)
- Custom report building
- Report scheduling and distribution
- Export to multiple formats (PDF, Excel, CSV)
- Report history and version control
- Interactive data visualizations

4.5.3 Generating a Report

To generate a report:

1. Navigate to the **Reporting System**

2. Select a report type:
   - Performance Report
   - Maintenance History
   - Alert Summary
   - Prediction Analysis
   - Custom Report
3. Configure report parameters:
   - Time period
   - Devices or components to include
   - Data points to analyze
   - Grouping and sorting options
4. Click **Generate Report**
5. View the report in the preview pane
6. Export the report using the **Export** button (PDF, Excel, CSV formats available)

Chapter5 Advanced Functionality

1 Anomaly Detection

5.1.1 Overview

The Anomaly Detection functionality utilizes advanced deep learning algorithms (such as Skip-GANomaly and PHOT) to identify non-normal patterns in engine operation. These anomalies may indicate potential failures or situations requiring maintenance.

5.1.2 Key Features

- Real-time anomaly detection
- Historical anomaly analysis
- Anomaly type classification
- Anomaly severity assessment
- Root cause analysis support
- Anomaly detection settings configuration

5.1.3 Understanding Anomaly Detection Results

Anomaly detection results are presented in several ways:

- **Anomaly Score:** A numerical value (0-100) indicating the degree of deviation from normal operation
- **Visual Indicators:** Color-coded highlights on dashboards and charts
- **Anomaly Details:** Information about the type and characteristics of detected anomalies
- **Temporal Context:** When the anomaly began and how it has evolved
- **Similar Cases:** References to similar anomalies detected in the past

2 Remaining Useful Life Prediction

5.2.1 Overview

The Remaining Useful Life (RUL) prediction functionality uses multiple deep learning models (such as CNN-LSTM, CNN-Transformer, DBN-BiGRU) to predict the remaining useful life of critical engine components. These predictions help maintenance teams plan maintenance activities at optimal times.

5.2.2 Key Features

- Component life prediction
- Prediction confidence intervals
- Multi-model prediction comparison

- Prediction trend analysis
- Maintenance window recommendations
- Prediction influencing factor analysis

5.2.3 Interpreting RUL Predictions

RUL prediction results provide valuable insights:

- **Estimated RUL:** The predicted time until maintenance is required
- **Confidence Interval:** The statistical range of the prediction
- **Degradation Curve:** Visualization of expected performance degradation
- **Influencing Factors:** Key parameters affecting the prediction
- **Recommended Actions:** Suggested maintenance activities and timing

3 Maintenance Schedule Optimization

5.3.1 Overview

The Maintenance Schedule Optimization functionality combines anomaly detection and life prediction results to help users develop optimal maintenance schedules. It considers multiple factors, including predicted failure risk, maintenance costs, downtime impact, and resource availability.

5.3.2 Key Features

- Maintenance task recommendations
- Maintenance window identification
- Maintenance resource allocation
- Scenario analysis (”what-if” simulations)
- Maintenance schedule conflict detection
- Cost-benefit analysis

5.3.3 Creating an Optimized Schedule

To create an optimized maintenance schedule:

1. Navigate to the **Maintenance Optimization** module
2. Configure optimization parameters:
    - Time horizon for planning
    - Available maintenance resources
    - Cost constraints
    - Operational requirements
    - Risk tolerance
3. Click **Generate Schedule**

4. Review the proposed schedule
5. Adjust parameters if needed
6. Save or export the optimized schedule

Chapter6 System Administration

1 User Management

6.1.1 Overview

The User Management functionality allows administrators to create and manage system user accounts, assign roles and permissions, and monitor user activity. This ensures proper access control and system security.

6.1.2 Key Features

- User creation and management
- Role definition and assignment
- Permission configuration
- Password policy management
- Access logs and auditing
- User activity reporting

6.1.3 Creating a New User

To create a new user account:

1. Navigate to **System Administration**
2. Select **User Management**
3. Click **Add User**
4. Complete the user information form:
   - Username
   - Password
   - Full name
   - Email address
   - Phone number
   - Role assignment
   - Department
5. Configure additional settings:
   - Account expiration (if applicable)
   - Password expiration policy
   - Multi-factor authentication requirements
   - Specific permissions overrides
6. Click **Create User**
7. The system will create the account and send login credentials to the specified email address

2 System Configuration

6.2.1 Overview

The System Configuration functionality allows administrators to customize system settings, including notification parameters, data retention policies, system integration settings, and global thresholds.

6.2.2 Key Features

- Global system settings
- Notification configuration
- Data management settings
- Integration settings
- Performance tuning
- Logging configuration

3 Data Management

6.3.1 Overview

The Data Management functionality provides tools for data storage, backup, archiving, and cleanup. It helps administrators maintain system performance and ensure data integrity and availability.

6.3.2 Key Features

- Data backup and recovery
- Data archiving and cleanup
- Data import and export
- Data quality monitoring
- Storage capacity management
- Data access control

Chapter7 Best Practices

1 System Usage Recommendations

- Regularly check the dashboard for system overview
- Configure custom alert thresholds for critical equipment
- Use multiple models for prediction comparison
- Regularly review and analyze anomaly detection results
- Compare system predictions with actual maintenance findings to optimize models
- Use the data simulation feature to test the impact of different scenarios
- Establish maintenance workflows that incorporate system recommendations
- Train operators on interpreting prediction results and uncertainty ranges

2 Data Interpretation Guidelines

- Understand prediction confidence intervals
- Focus on trends rather than individual data points
- Associate anomaly patterns with known issue types
- Consider the impact of operating conditions on predictions
- Use correlation analysis to identify influencing factors
- Compare prediction results with historical patterns
- Consider multiple data sources when making maintenance decisions
- Document findings and outcomes to improve future analyses

3 Troubleshooting Tips

- Ensure sensor data quality and connectivity
- Verify input parameters and configuration settings
- Check system notification and alert configurations
- Use data simulation to validate issues
- Review system logs for error messages
- Contact the technical support team for advanced assistance
- Check for recent system updates or changes that may affect behavior
- Verify component configuration information is current and accurate

Chapter8 Appendices

1 Frequently Asked Questions

1. **How often is sensor data collected?**
   Sensor data collection frequency varies by component type and operational context, but typically ranges from once per second for critical parameters to once per minute for standard monitoring.
2. **What does the confidence score mean in predictions?**
   The confidence score (0-100%) indicates the statistical reliability of the prediction based on available data and model accuracy. Higher scores indicate greater certainty in the prediction.
3. **How are anomalies detected?**
   Anomalies are detected using advanced deep learning models (Skip-GANomaly and PHOT) that learn normal operation patterns and identify deviations from these patterns.
4. **Can the system integrate with our existing maintenance management software?**
   Yes, the system provides API integrations for common maintenance management systems. Contact your system administrator for specific integration options.
5. **How often should models be retrained?**
   Models should be retrained quarterly or when performance metrics indicate degradation in prediction accuracy. The system administrator can schedule automated model evaluation and retraining.

2 Error Message Reference

Error Code	**Description and Resolution**
ERR-1001	Database connection failure. Contact system administrator to verify database service status.
ERR-1002	Authentication failure. Verify your username and password, or request a password reset.
ERR-2001	Sensor data collection error. Check sensor connectivity and configuration.
ERR-2002	Data processing timeout. The operation took too long to complete. Try again with a smaller data set.
ERR-3001	Model inference error. The prediction model encountered an issue. Contact technical support.
ERR-3002	Insufficient data for prediction. More operational data is required for accurate prediction.
ERR-4001	Report generation failure. Verify report parameters and try again.
ERR-5001	Permission denied. You do not have sufficient privileges for this operation. Contact your administrator.

3 Related Documentation

- System Design Document

- Installation and Configuration Guide
- API Reference Manual
- Data Dictionary
- Training Materials
- Maintenance Manual
- Security Compliance Guide
- Integration Specifications

Chapter9　Copyright and Legal Notice